Name: ____________________________ Date: ________________ Period: _________

1. For the rational function $r(x) = \dfrac{5}{(x-3)(x+1)}$, find all vertical asymptotes. Write each as an equation.

   Vertical asymptote(s): $x = 3$ **and** $x = -1$

   (Denominator zeros: $x - 3 = 0 \Rightarrow x = 3$ *and* $x + 1 = 0 \Rightarrow x = -1$*. Numerator is 5, nonzero at both. Both are VAs.)*

2. A table of values near $x = 2$ is shown below.

x	1.9	1.99	2.001	2.01	2.1
$r(x)$	47	497	501	51	5.7

   Based on the table, what does the behavior of $r(x)$ as $x \to 2$ indicate about $x = 2$? Circle the best answer and explain briefly.

   A. Hole at $x = 2$ **B.** **Vertical asymptote at** $x = 2$ **C.** Neither

   Explanation: **The outputs grow without bound ($\to \infty$) as $x \to 2$ from both sides, indicating a vertical asymptote, not a hole. A hole would produce outputs approaching a finite value L.**

3. Factor the expression $x^2 + x - 6$ completely.

   $x^2 + x - 6 = (x+3)(x-2)$